

DATADOG ON OPENSIFT: MASTER TECHNICAL BLUEPRINT

Advanced Architecture, LLD, and Operator-Led
Deployment Guide for OpenShift 4.x.

[TARGET: OCP 4.x]

[MODE: EXPERT]

[RUNTIME: CRI-O]

Executive Summary & Telemetry Goals



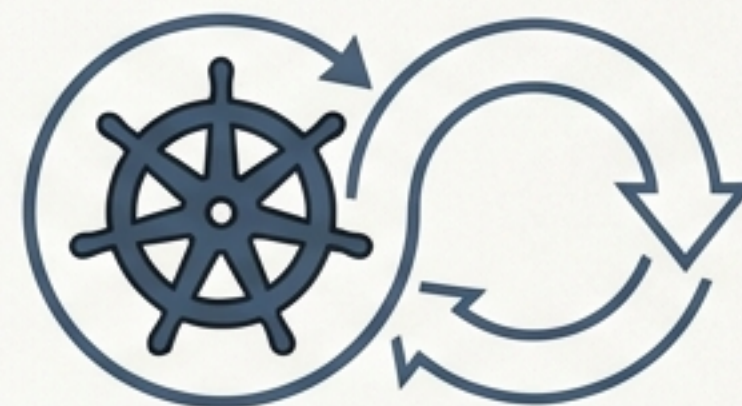
360-Degree Multi-Tenant Visibility

Achieving deep observability across **node**, **cluster**, and **application** layers without requiring manual instrumentation per team.



Strict Security Compliance

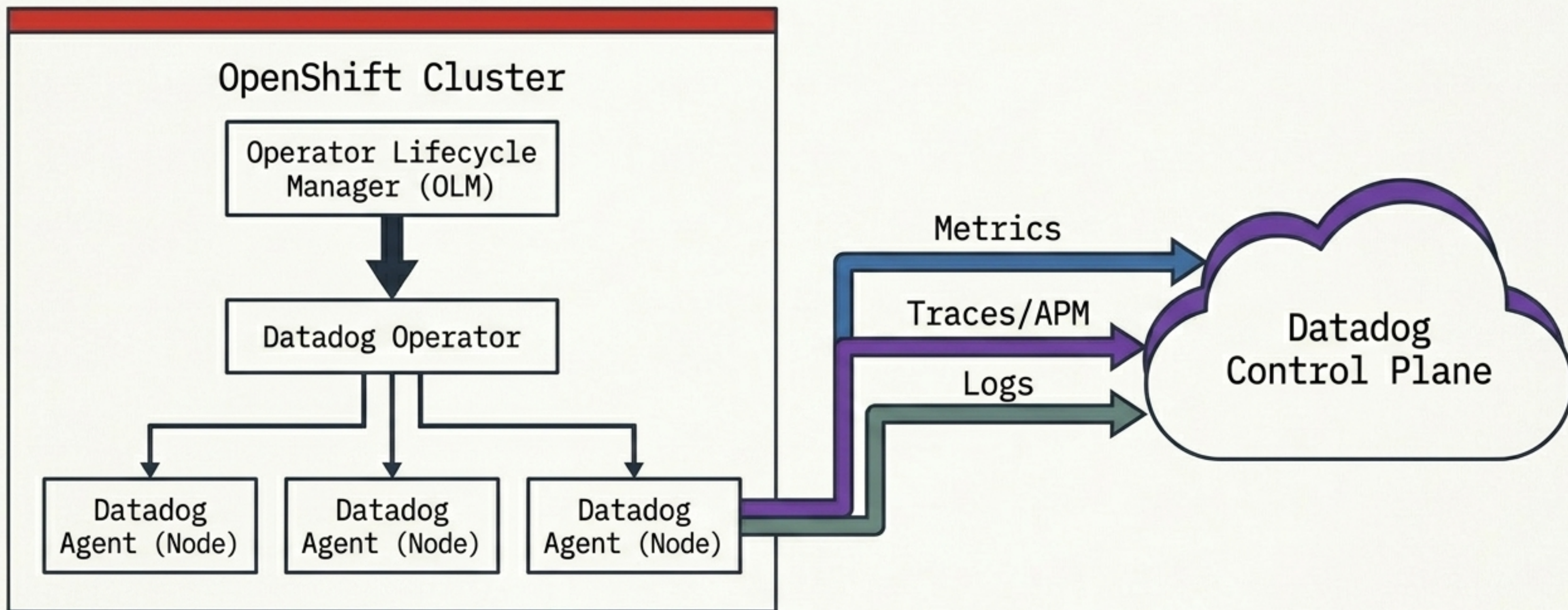
Bypassing generic limitations of OpenShift's **restricted-v2** defaults securely by implementing custom **Security Context Constraints** (SCCs).



Modern Lifecycle Management

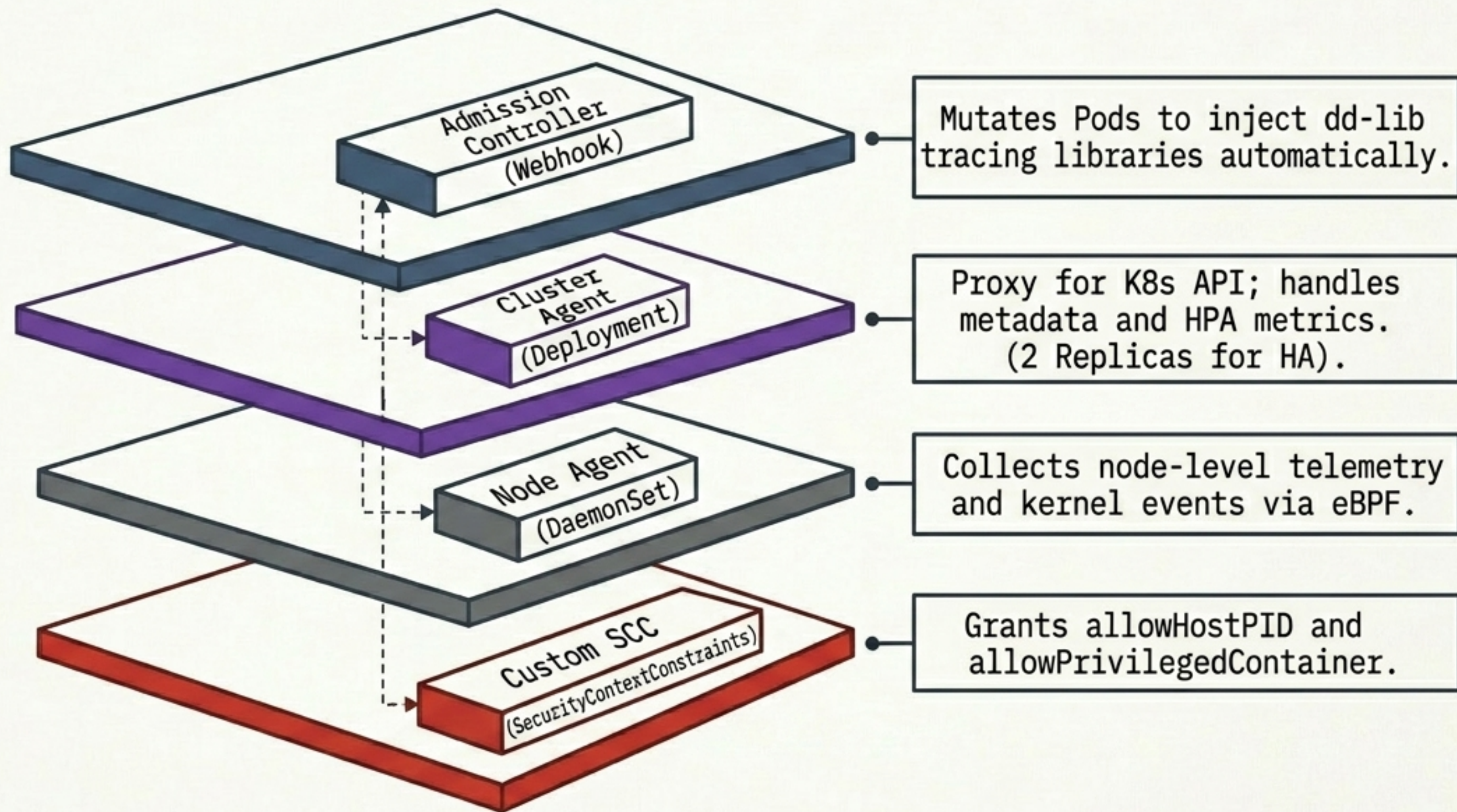
Moving away from static **DaemonSets** to a native **Kubernetes** reconciliation loop via **Operator Lifecycle Manager** (OLM).

High-Level Architecture (HLD)



Operator-led lifecycle ensures the agent configuration dynamically mirrors Datadog's best-practice defaults and K8s API reconciliation.

Low-Level Architecture (LLD)



Platform Engineering Object Mapping

Datadog Component		OCP Primitive	Key Responsibility	Required Privilege
Node Agent		DaemonSet	eBPF & host logs	Needs hostip & SCC custom access
Cluster Agent		Deployment	API Metadata & HPA	RBAC to read K8s API
APM Injector		MutatingWebhookConfiguration	Library Injection	Intercepts Pod creation
Security Policy		SecurityContextConstraints	Unlocks deep observability	Needs cluster-admin to apply

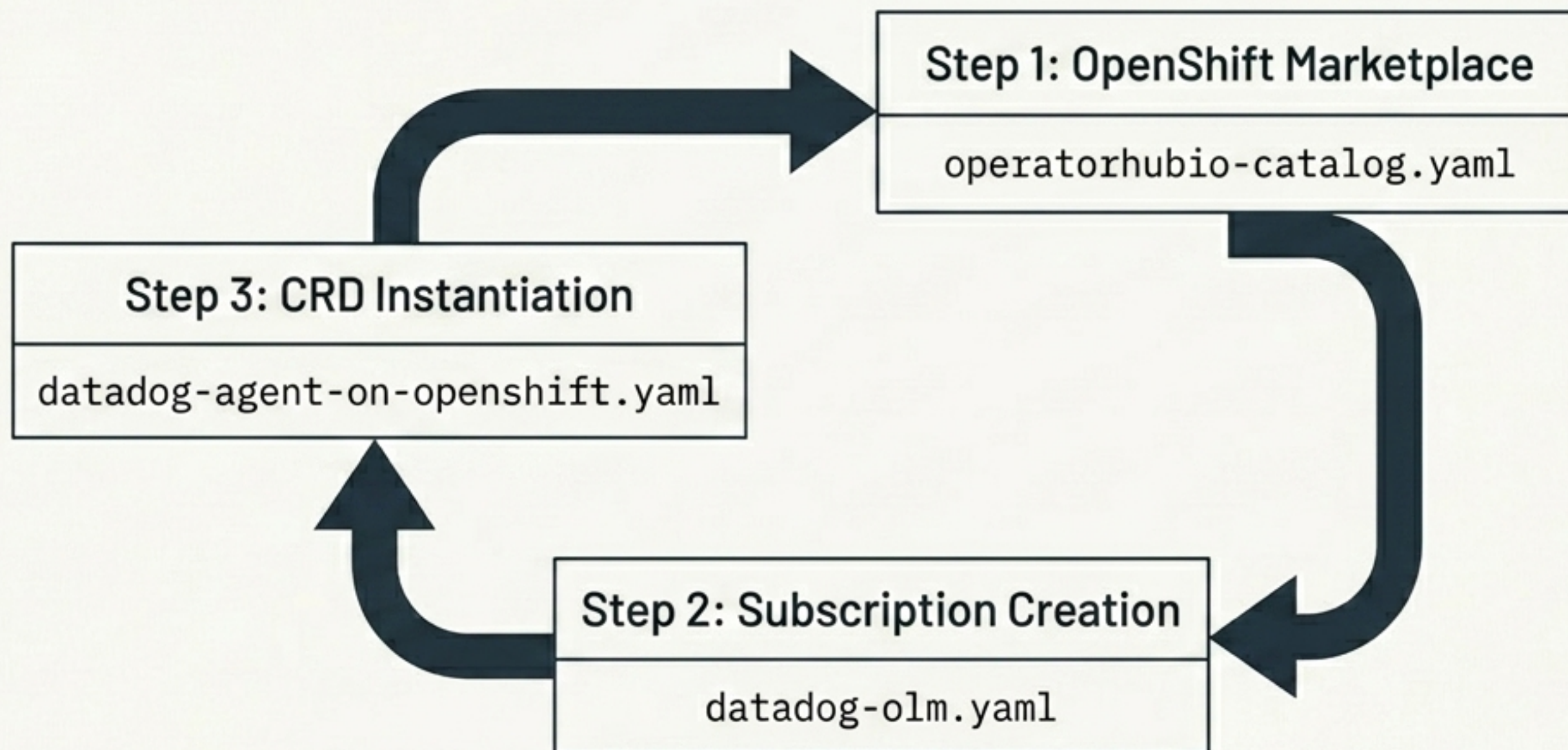
Every abstraction maps to a tangible OpenShift requiring explicit RBAC and SCC clearance.

Architectural Comparison Matrix

	Solution 1 (Helm v3)	Solution 2 (Operator/OLM - Recommended)	Solution 3 (Manual DaemonSet)
Lifecycle Management	Manual upgrades	K8s Reconciliation Loop ✓	High error rate ✗
K8s API Native Integration	Standard	First-class CRD resource ✓	None ✗
Default Best Practices	Static values.yaml	Dynamic built-in defaults ✓	User burden ✗

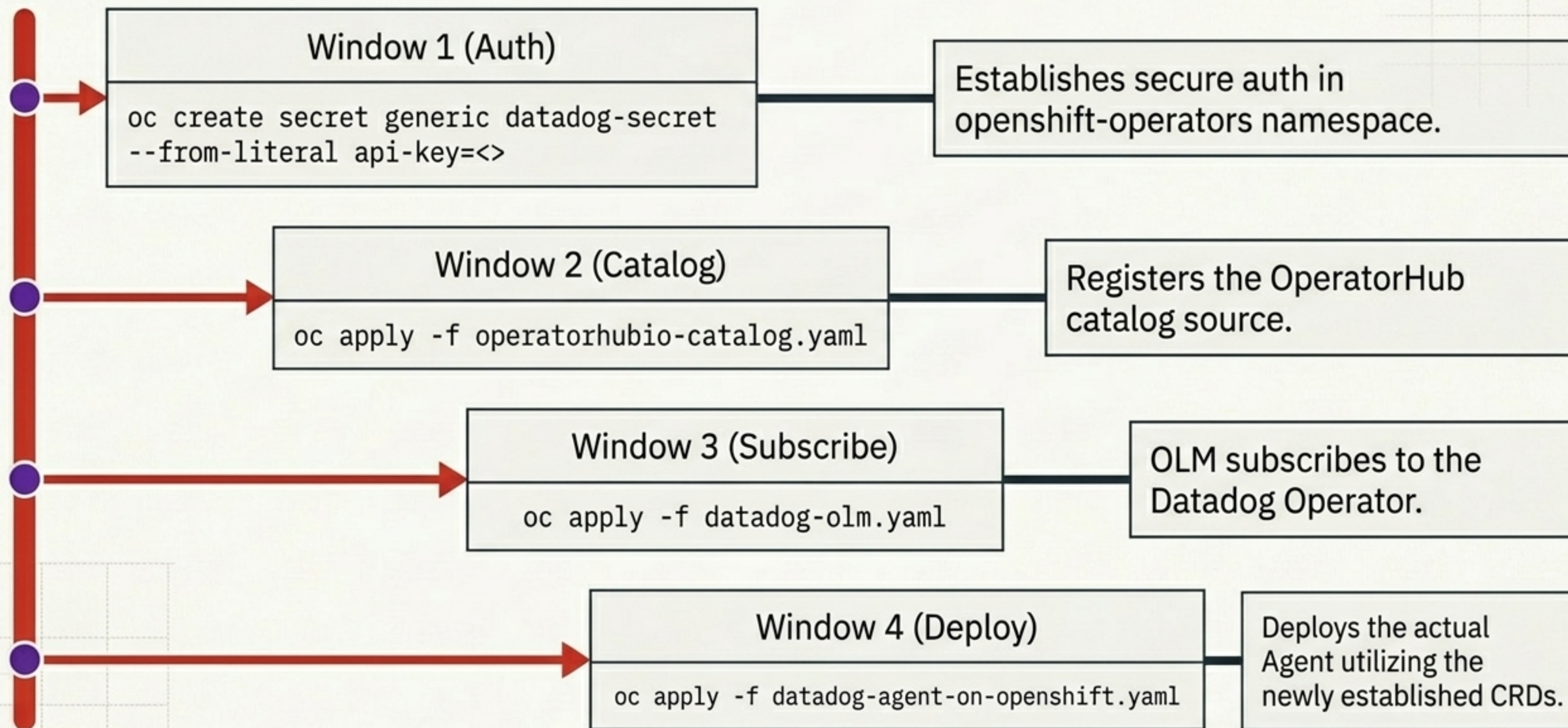
The Operator pattern (Solution 2) is the Enterprise standard, seamlessly integrating Datadog into the K8s reconciliation loop.

The Recommended Path (Solution 2 - OLM)



By utilizing OLM, default SCCs are automatically associated with the datadog-agent-scc ServiceAccount, streamlining the security binding process.

Step-by-Step Installation (Day 1)



OpenShift Security Context Constraints (SCC)

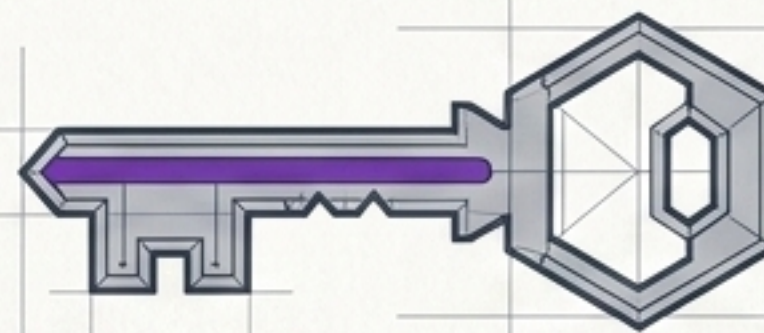
The Problem



The default restricted-v2 OpenShift SCC blocks binding to the host port.

However, hostip mode is strictly required for OpenShift APM to ensure a constant, discoverable target IP.

The Solution

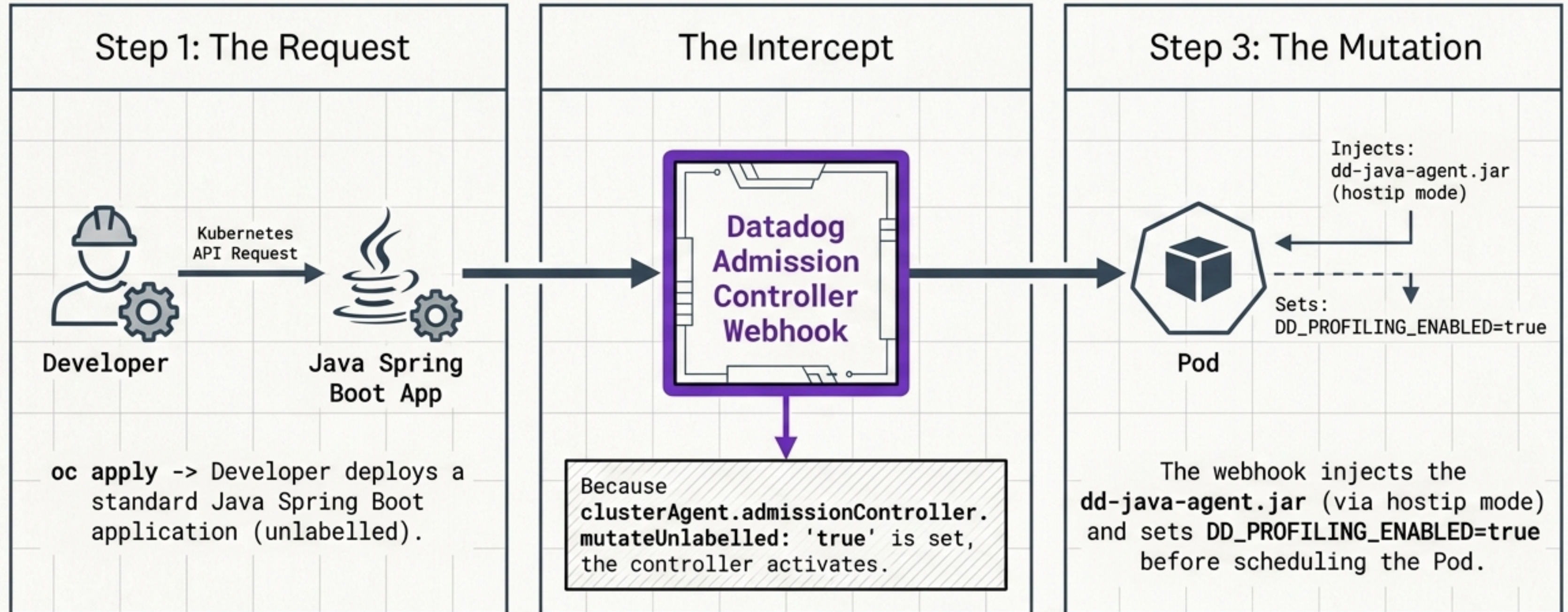


scc.yaml

```
allowHostDirVolumePlugin: true
allowHostNetwork: true
allowHostPID: true
allowPrivilegedContainer: false
```

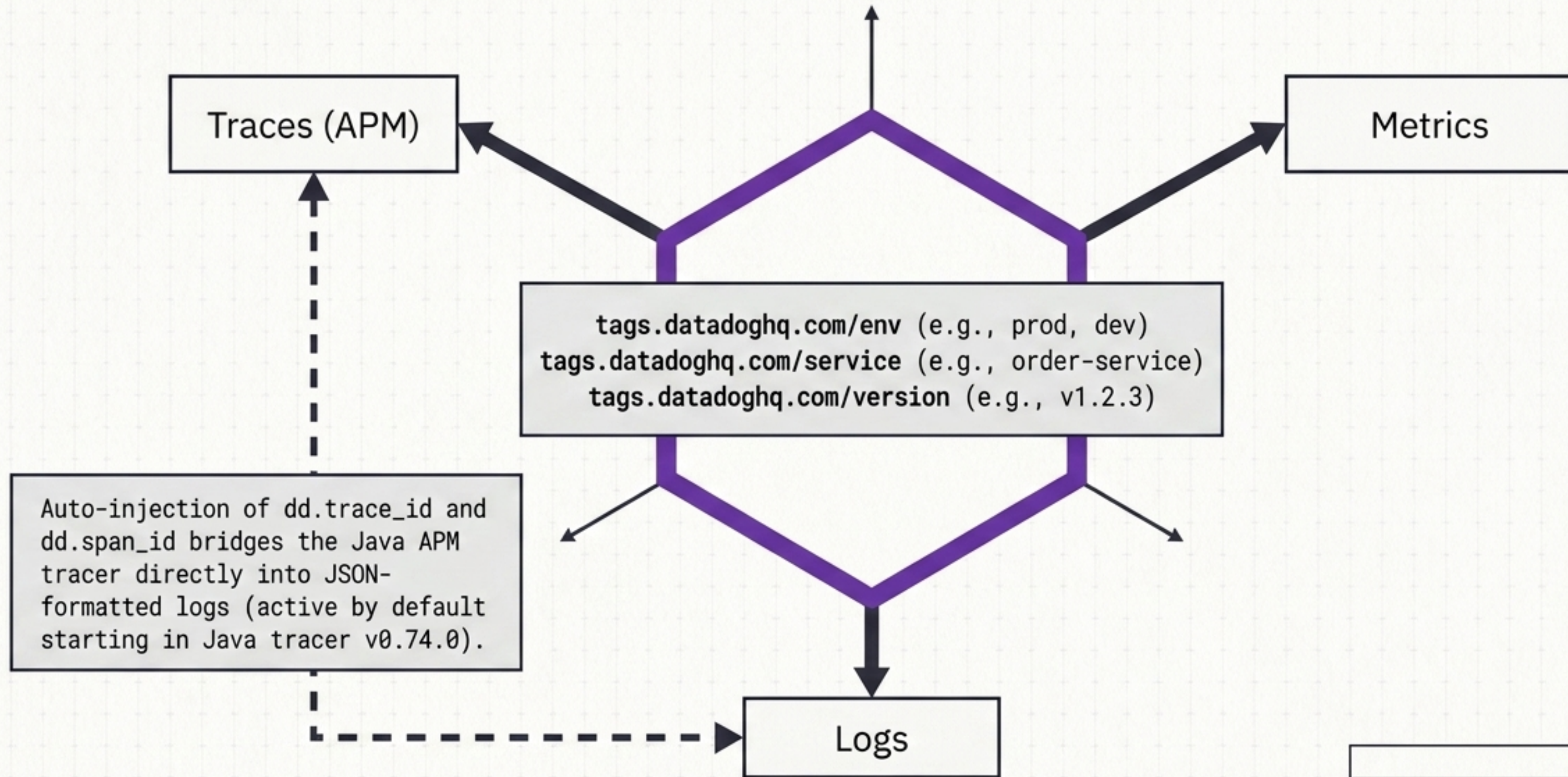
Applying this custom SCC directly to the Datadog ServiceAccount prevents the need to alter cluster-wide default constraints.

APM & The Mutation Flow



Result: Zero-touch APM instrumentation for development teams.

Unified Tagging & Correlation



Advanced Autodiscovery (JMX & Prometheus)

JMX Autodiscovery

Required Pod Annotations to trigger standard Java checks.

```
ad.datadoghq.com/<CONTAINER_NAME>.check_names:  
'["<INTEGRATION_NAME>"]'
```

Note: Used for standard **Java runtime metrics** discovery.

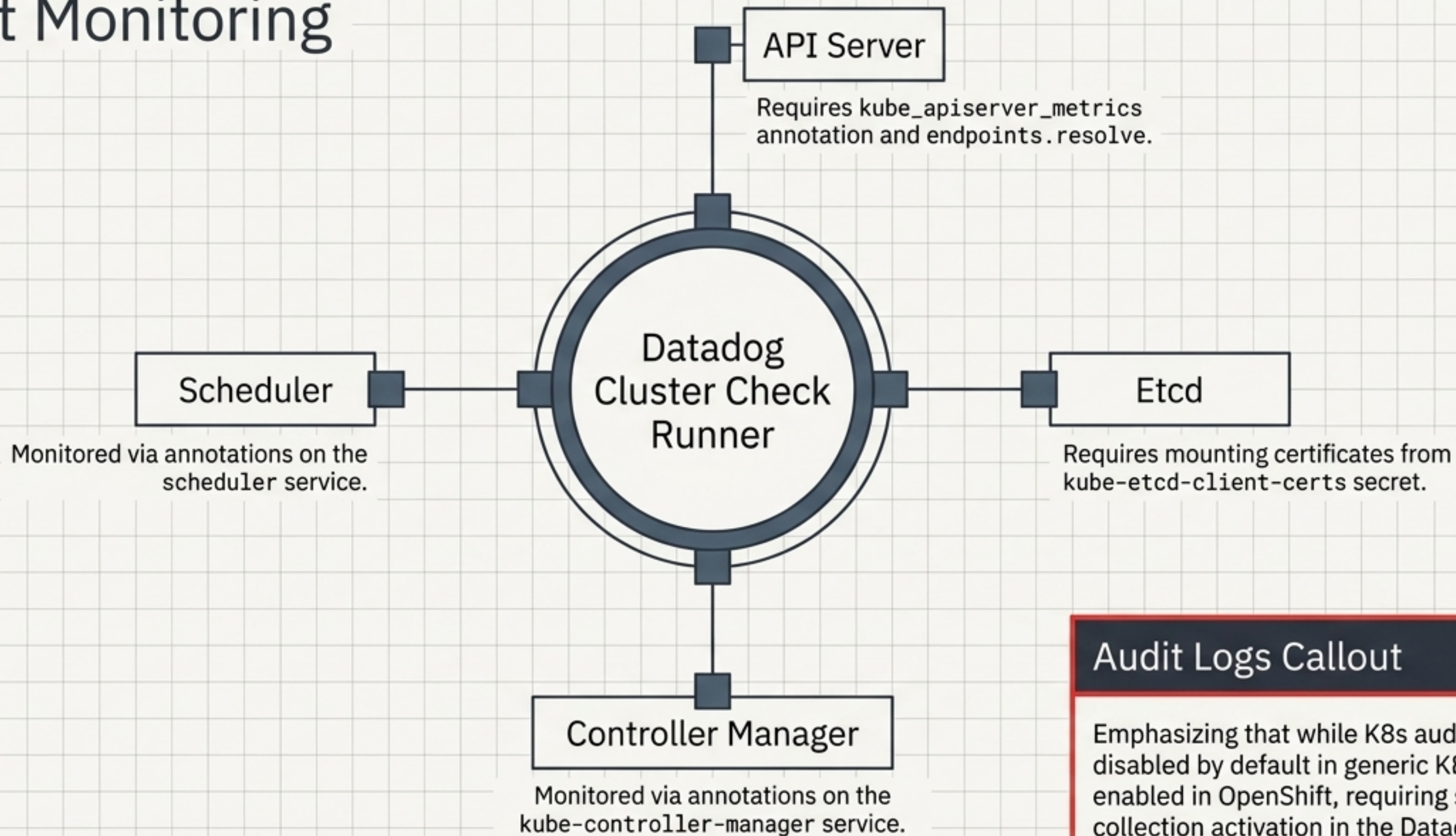
Prometheus Scrape (prometheusScrape)

Configuration for scraping custom endpoints like Tibco BWCE.

```
features.prometheusScrape.enabled: true
```

Allows scraping of **Custom Prometheus Metrics** exposed on **Pod port 9095**.

Kubernetes Control Plane & Audit Monitoring



Audit Logs Callout

Emphasizing that while K8s audit logs are disabled by default in generic K8s, they are enabled in OpenShift, requiring specific log collection activation in the Datadog Agent.

Resource Profiles & FinOps

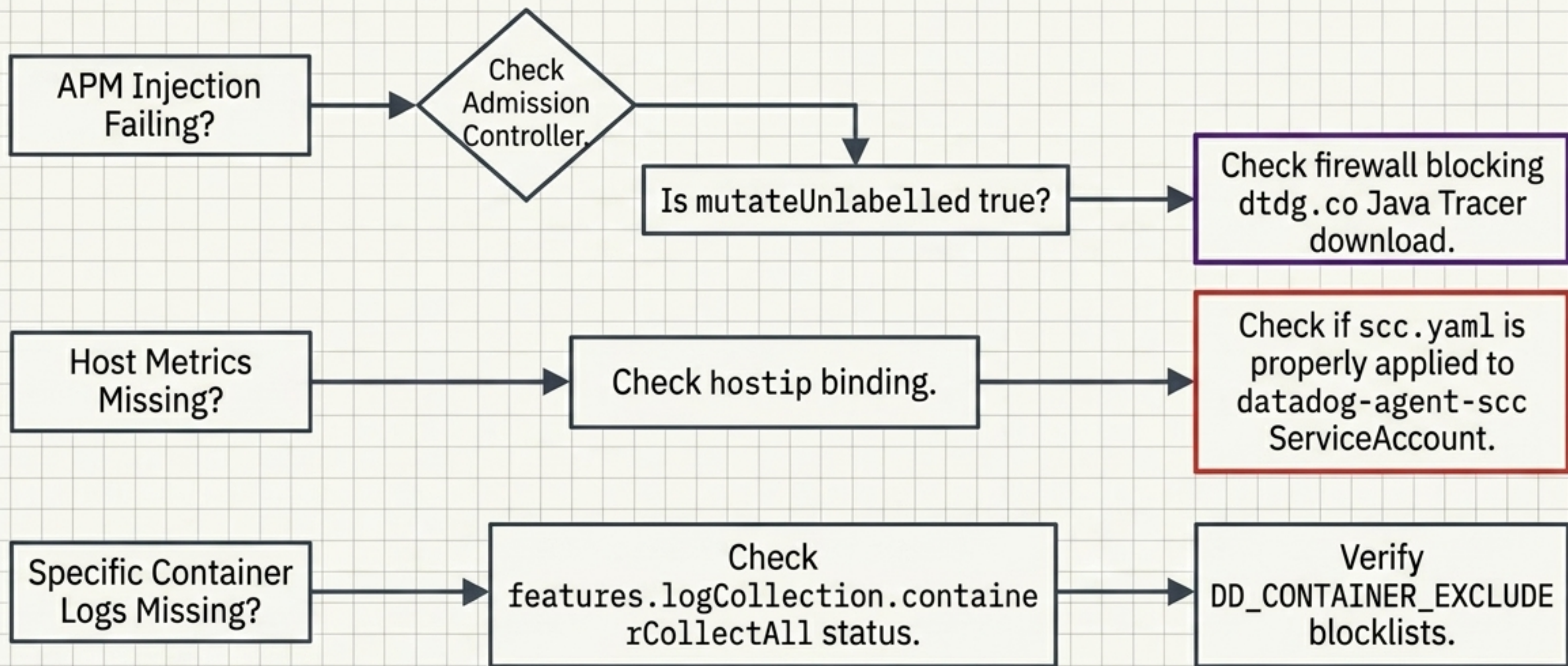
Resource Profiling Table			
Component	CPU (Req/Lim)	RAM (Req/Lim)	Scale
Node Agent	200m / 500m	256Mi / 1Gi	Per Node
Cluster Agent	200m / 500m	256Mi / 512Mi	HA (2 Replicas)
System Probe (eBPF)	100m / 300m	128Mi / 512Mi	Per Node

FinOps Data Flow



Usage Attribution based on env and service tags to track APM/USM costs per team.

Troubleshooting Decision Tree



Reference the **Master Engineering Documentation** (Sections 18-20) for complete diagnostic tracing commands.